INCH-POUND

MIL-P-63140A (AR)
AMENDMENT 3
5 JAN 1989
SUPERSEDING
AMENDMENT 2
25 OCTOBER 1985

MILITARY SPECIFICATION

PROJECTILE 8-INCH HE, RA M650
DELAY ASSEMBLY FOR

This amendment forms a part of Military Specification MIL-P-63140A (AR), dated 24 March 1980 and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1 Under Military Standards, change:

"MIL-STD-453 to MIL-STD-00453"

Add the following Military Standard:

"MIL-STD-410 - Nondestructive Testing Personnel Qualification and Certification"

PAGE 2

2.1 Inspection Equipment Drawings:

Delete: "9325984 - Envelope Drawing (Porosity Test)" and substitute: "9246420 - Envelope Drawing (Porosity Test)"

Delete: "9325985 - Nitrogen Test Assembly" and substitute: "9326889 - Nitrogen Test Assembly"

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3.3.1 Delete in its entirety

* 3.4 Delete in its entirety and substitute the following:

"3.4 Crimp strength. The washer crimp shall be capable of withstanding a 225 pound minimum load when tested as specified in 4.4.3.6."

3.7 Flight test. Delete in its entirety and substitute the following:

"3.7 Flight test.

3.7.1 Delay chemicals acceptance.

3.7.1.1 Pilot mixtures. Small batches from each new lot of chemicals shall be prepared as shown in Table II.

TABLE II

Pilot mixtures

Batch #	%W	%$BaCrO_4$	%$KClO_4$
1	45.00 ± .10	40.50 ± .10	14.50 ± .10
2	46.00 ± .10	39.50 ± .10	14.50 ± .10
3	47.00 ± .10	38.50 ± .10	14.50 ± .10
4	48.00 ± .10	37.50 ± .10	14.50 ± .10
5	49.00 ± .10	36.50 ± .10	14.50 ± .10

% VAAR shall be kept constant at approx. 1%.

3.7.1.2 Testing. Ten delay assemblies from each batch shall be submitted for ballistic firing. After conditioning at 70°F for a period of 24 hours minimum, the delays shall be flight tested in accordance with 4.5.3.1.1.

3.7.1.3 Tungsten curve. The ballistic results shall be plotted against the percent tungsten. From this curve the optimum tungsten percent will be determined. The chemical composition is optimized at a delay time of 7.8 seconds.

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3.7.1.4 Usage. The results from paragraphs 3.7.1.1 - 3.7.1.3 shall be used in subsequent delay composition preparation.

3.7.2 Delay composition acceptance.

3.7.2.1 Plus 70°F. After conditioning at +70°F for a period of 24 hours min., the delay shall be flight tested at a proving ground as specified in 4.4.3.7.1.1. The average delay time shall be 7.8 plus or minus .75 sec., and the standard deviation of the delay times shall not exceed 0.5 sec.

3.7.2.2 Plus 145°F. After conditioning at +145°F for a period of 24 hours min., the delay shall be flight tested at a proving ground as specified in 4.4.3.7.1.2. The average delay time shall be 6.8 plus or minus .75 sec. and the standard deviation of the delay times shall not exceed 0.5 sec.

3.7.3 Delay assembly acceptance.

3.7.3.1 Plus 145°F. After conditioning at +145°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.7.2.1). The average delay time shall be 6.80 seconds plus or minus .75 seconds, and the standard deviation shall be no greater than .5 sec.

3.7.3.2 Minus 50°F. After conditioning at minus 50°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.7.2.2). The average delay time shall be 9.30 seconds plus or minus .75 seconds, and the standard deviation shall be no greater than .8 sec.

3.7.3.3 Plus 70°F. After conditioning at +70°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.4.3.7.2.3). The average delay time shall be 7.8 seconds plus or minus 0.5 seconds and the standard deviation of the delay times shall be no greater than 0.5 seconds."

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Add the following new subparagraph:

"3.10 Process control for delay increment consolidation. A process control description shall be developed for delay increment consolidation, and submitted for government approval. The description shall include all elements of press setup, operation, calibration, maintenance, and operator instruction required to assure that the required number of delay increments are individually consolidated at the required pressure and for the required minimum consolidation time. The process controls shall include verification that the required pressure and dwell are met at the extremes of the press consolidation range (Section 4.4.3.1 and 6.12)."

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4.4.2.4 Delete Minor Defect 201

PAGE 12

4.4.2.5 Add the following Minor Defect "203 Distance tube end surface to inside bottom of igniter plug, min. -, 0.65%, 3.2, Gage".

PAGE 13

4.4.3.1 Delete in its entirety and substitute the following:

"4.4.3.1 Consolidation load. Critical Defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. Proper calibration of each consolidation machine shall be verified at the beginning, middle and end of each production shift. Dwell time and consolidation load shall be measured with a stopwatch and a load cell. If any consolidation machine fails verification for dwell time, consolidation load or for not maintaining proper load through the dwell time, all the delays manufactured on that machine since the last acceptable verification shall be rejected. In addition, a warning device indicating failure of dwell time and failure of consolidation load shall be assembled to each consolidation machine. The warning device or mechanism shall provide an easily detected signal to the machine operator. If during loading of the delay the consolidation load or dwell time requirements are not met, that delay shall be rejected."

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4.4.3.7.1.1: Change reference paragraph 3.7.1.1 to 3.7.2.1.

Add the following new paragraph:

"4.4.3.7.1.2 Plus 145°F - Major defect. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the Government for flight testing. If the requirement of Paragraph 3.7.2.2 is not met - Major defect, or if the delay assembly lot fails to ignite - Critical defect, or ignites in the tube - Critical defect, the delay assembly lot shall be rejected. The test shall be conducted in accordance with Paragraph 4.5.3.1.2. A minimum of 8 radar delay times shall be required to determine delay time acceptance."

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4.4.3.7.2.1 and 4.4.3.7.2.2: Delete in their entirety and substitute the following paragraphs:

4.4.3.7.2.1 Plus 145°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.7.3.1 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.2.1.

4.4.3.7.2.2 Minus 50°F. A sample of ten (10) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.7.3.2 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.2.2".

4.4.3.7.2.3: Change reference paragraph 3.7.2.3 to 3.7.3.3.

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4.4.3.8: Add the following at the end of the paragraph:

"Should the solids content be too high, additional solvent may be added, with agitation, until the solids content is acceptable. The VAAR solution will then be retested to assure compliance."

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4.5.1: Delete the first sentence and substitute the following:

"Each delay assembly shall be conditioned at plus 145°F and assembled to the porosity test equipment."

PAGE 17

4.5.2: Delete the last sentence and add the following at the end of the paragraph:

"In lieu of a penitrometer, a radiographic standard shall be manufactured to assure adequate sensitivity of the radiographic technique. A delay will be manufactured with a low density pocket in each of the two delay columns. Normal production materials shall be used to create the standard delay. In both columns the flash and first delay increment shall be pressed in correctly. In one column the second and third delay increments shall be placed in the column but not pressed, the fourth and fifth increments shall be pressed. In the other column the second increment will be correctly pressed, but the third and fourth increments will be loose loaded, the fifth increment will be pressed. Both columns will be back drilled, have igniter pressed in and have washers assembled. The delay manufactured as above will have a void between the first and second delay increments in one column and a void between the second and third increments in the other column. These voids must be clearly visible when the standard delay is radiographed. Using the standard delay, film sensitivity shall be checked on each piece of film used for delay acceptance. Photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.2.1."

Add the following new subparagraph:

"4.5.2.2 Operator qualification. For operation of radiograph equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level I. For set-up and calibration of radiographic equipment, personnel shall be qualified and certified in accordance with MIL-STD-410, Level II."

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6.8: Delete in its entirety and substitute the following:

"6.8 Pilot mixture. Composition for use in M650 delays may also be qualified by the pilot mixture tests prescribed in MIL-P-50570 (M549 Delay). This one pilot mixture test may qualify composition for both M549 and M650 delay assemblies."

PAGE 21

Add the following new subparagraph:

"6.12 Submission of process control description. The process control description for delay increment consolidation shall be prepared in accordance with DI-E-10392 and submitted to the responsible government technical agency for approval.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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